PYTHON CASE STUDIES — EASY EXPLANATION

Quick one-page notes • Variables • Operators • Conditions • Loops • Lists

1. Progressive Electricity Billing

Goal: Calculate an electricity bill from the number of units used. Different ranges of units may have different per-unit rates (called **slabs**). Then add any fixed charge and apply a surcharge or rebate if required.

Simple logic

```
Read units
bill = 0
For each slab (in order):
find units that fall inside that slab
bill = bill + (units_in_slab × slab_rate)
bill = bill + fixed_charge
If surcharge applies: add it; otherwise, if rebate applies: subtract it
Display the final bill
```

Remember: Use **if/elif/else** for conditions, a **loop** to process slabs, and variables to store units and the running total. Use the exact slab rates and rules given in your question.

2. Railway Fare & Reservation Calculator

Goal: Find the ticket total using passenger details, journey distance, class, age concession, reservation fee and cancellation penalty (if applicable).

Simple logic

```
Read passenger details, distance, class and booking/cancellation choice
fare = distance × rate_for_selected_class
If passenger qualifies for age concession: reduce fare
Add reservation fee
If cancelled: calculate and add/apply cancellation penalty as specified
Display the final amount
```

Remember: Use **if/elif/else** to select class and concession rules. A **list** can store several passengers or their fares, and a **loop** can calculate each passenger's total.

Python concepts used

Concept	Easy meaning	Used for
Variables	Named storage for values	units, fare, bill, total
Operators	+ – × comparisons	calculations and checking rules
Conditions	Choose what happens next	slabs, class, concession, cancellation
Loops / Lists	Repeat steps / store many values	slabs or multiple passengers

Exam tip: Read the question carefully, write the inputs first, calculate step by step, check every condition, and print a clear final total. Do not guess rates or concession percentages—use the values provided.